

Algebra 1
Unit 13
Lesson H2 Practice

Name _____
Date _____
Period _____

Use the functions $b(x) = -2x^2 + 6x - 3$ and $c(x) = 8 + 3x$ to determine each new function. Write your answer in standard form.

1. $(b + c)(x)$

2. $(b - c)(x)$

3. $(c - b)(x)$

4. $(b \cdot c)(x)$

5. $\left(\frac{b}{c}\right)(x)$

6. Write the restriction on the domain for $\left(\frac{b}{c}\right)(x)$.



Paulina and Stefan are renting a boat for a trip to the Keys. The boat costs \$200 per day. The rental includes 125 miles for free and then charges \$0.50 per mile. The function $B(m) = 0.50(m - 125) + 200$ models the cost of the boat, B , in terms of miles traveled, m . They determine gas will cost \$1.75 for each mile they travel. The company charges a \$75 fee for the first tank of gas. The function $F(m) = 1.75m + 75$ models the cost of gasoline for their vacation.

7. What is a reasonable domain for $B(m)$?

8. What is a reasonable domain for $F(m)$?

9. Determine the function that gives the total cost, $C(m)$, in dollars, based on the miles traveled.

10. What is a reasonable domain for $C(m)$?